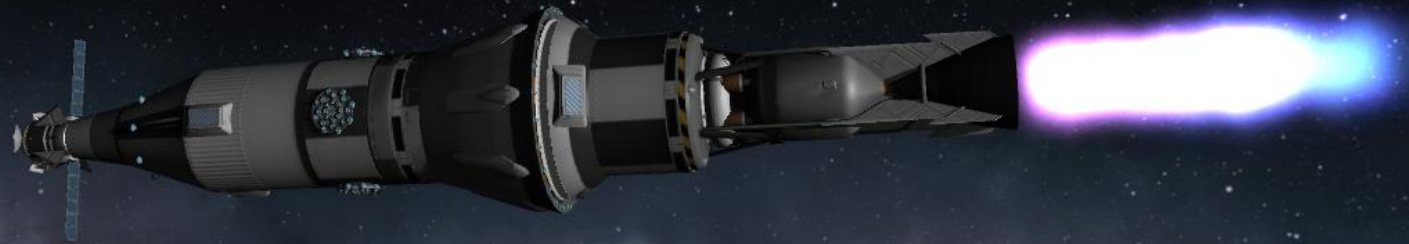


CFD Simulation of Hydrogen Propellant Flow in Nuclear Thermal Propulsion Reactor Cores



Overview

- Chapter 1: Overview & History of nuclear propulsion development to contextualize NTR examined
- Chapter 2: Overview of historical modeling tools & methods and tools used in this work
- Chapter 3: Theoretical overview of how the PIMPLE algorithm operates in this setup
- Chapter 4: Detailed Overview of reactor core anatomy down to the single fuel cluster
- Chapter 5: Description of all OpenFOAM v13 model parameters for single equivalent channel in the core center region
- Chapter 6: Exposition and discussion of Results with validation and mesh sensitivity study.
- Chapter 7: Overview of how minimal cluster-scale validation model can be realized
- Chapter 8: Conclusion with pipeline for further validation and application
- Appendices: Curated & contextualized information relevant for qualification and reproducibility

Chapter 1: Motivation for Nuclear Propulsion

Tsiolkovsky's Rocket Equation:

$$\Delta v = v_e \ln \left(\frac{m_0}{m_1} \right) \quad ISP[s] = \frac{v_e \left[\frac{m}{s} \right]}{g \left[\frac{m}{s^2} \right]}$$
$$\Delta v \propto v_e$$

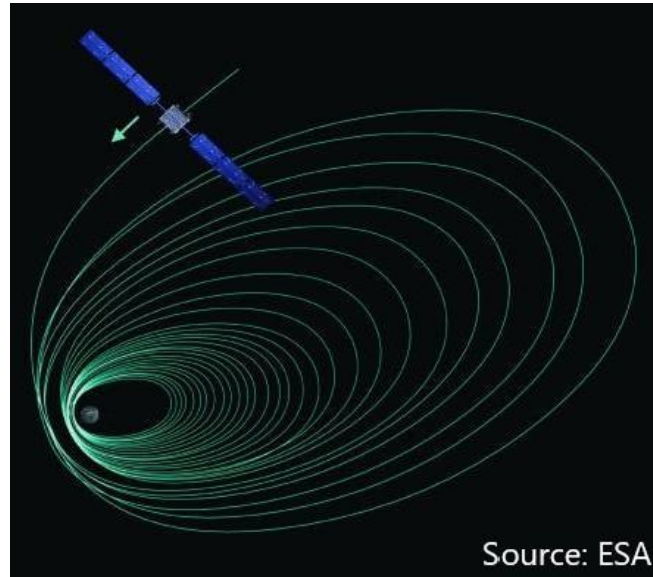
"Tyranny of the Rocket Equation"

$$\Delta v = v_e \ln \left(\frac{m_{empty} + m_{payload} + m_{propellant}}{m_{empty} + m_{payload}} \right)$$

$$m_{propellant} = \left(\exp \left[\frac{\Delta v}{v_e} \right] - 1 \right) \cdot (m_{empty} + m_{payload})$$

$$m_{propellant} \propto \exp[\Delta v]$$

Additional constraint for crewed flight: Thrust

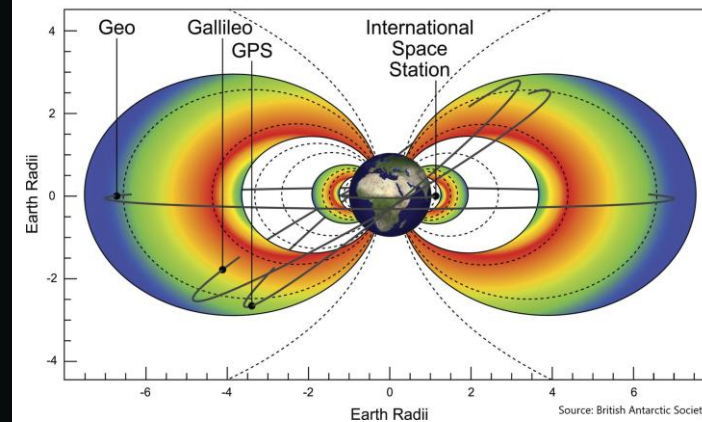


Source: ESA

Low-thrust maneuvers:

- Efficient but time-consuming
- To minimize crew dose, flight duration should be minimal.
- Timely orbital ejection requires sufficient initial thrust/mass

Satellite orbits and the van Allen radiation belts



Van Allen Belts:

- Capture and bundle Solar particle radiation
- Compounded by Solar storms

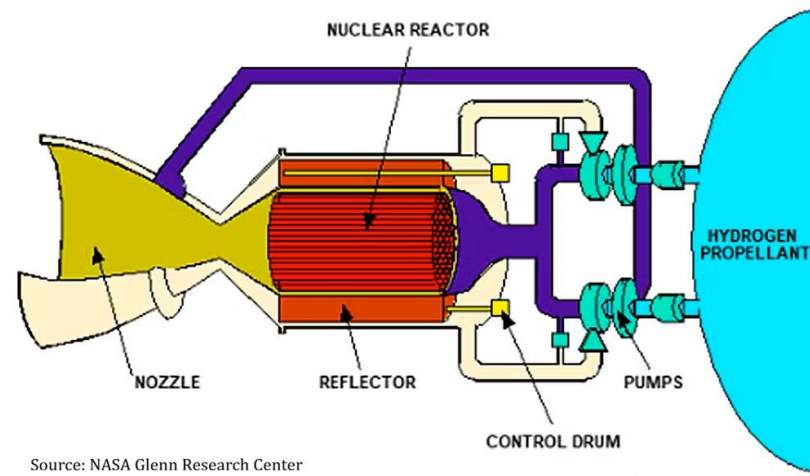
Chapter 1: Overview of Nuclear Propulsion Methods – Comparison of developed Propulsion Systems

Hydrogen-Oxygen Rocket



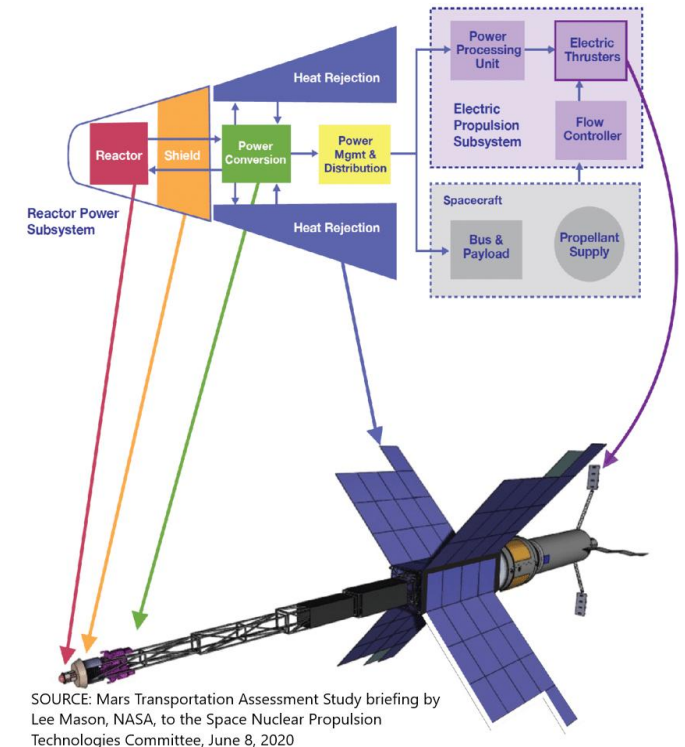
- Specific impulse 450s
- Most propellant-efficient chem. propulsion
- High TWR (can lift off Earth)
- Limited by chemical energy density

Nuclear Thermal Propulsion



- Specific impulse 800-900s
- ca. 1-2h burn time
- Hundreds of kN of thrust for 1GW
- Matured to practical flight-readiness
- Structural integrity of lightweight open-cycle reactor core must be guaranteed
- Fuel corrosion by high-temperature hydrogen must be minimized

Nuclear Electric Propulsion



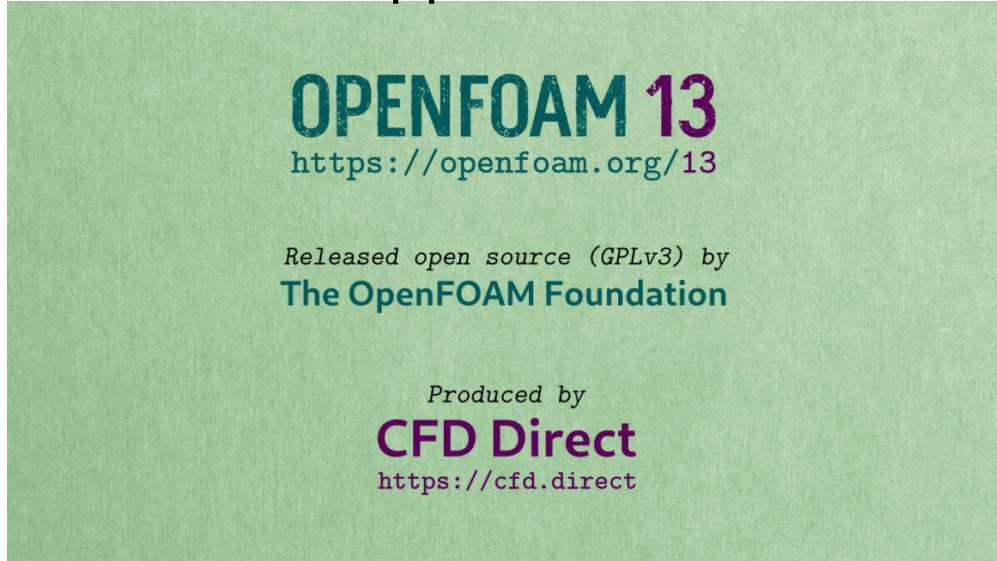
- Specific impulse 1000-20'000s
- Long service life (years)
- Low thrust (ca. 1mN-1N per 1kWe)
- Power conversion losses: thermal → electrical → kinetic
- Ample heat rejection necessary

Chapter 1: Chronicle of Nuclear Space Propulsion Development

- 1957-1965: Project Orion theorizes nuclear pulsed propulsion
 - Concept tested atmospherically with chemical explosives
- 1965: SNAPSHOT Mission launches SNAP-10A experimental nuclear electric propulsion satellite
 - First electric propulsion in space (tested on battery power)
 - First nuclear reactor power generation in space (590W max)
- 1955-1973: Rover Program develops flight-ready NTR under NERVA project
- 1970s-1980s: USSR develops RD-0410 NTR
 - Fuel corrosion decreased by adding hexane to hydrogen propellant
 - Compared to NERVA: Less thrust but higher efficiency
- 2009-today: Roscosmos & Rosatom commission TEM nuclear-electric space tug of Megawatt class
 - Intended for first uncrewed mission to outer planet (Jupiter) and 10-year service life
 - Slow but ongoing development (droplet radiator, Hall-effect thrusters, HTGR)
- 2021-2025: DARPA & NASA commission DRACO HALEU-fueled NTR (cancelled)
 - Ca. 20% enrichment
 - Orbital test flight originally planned for 2027

Chapter 1: Problem Description

The Application...



- Comprehensive open-source CFD application
- Many functionalities
- Built on C++ code

...to be validated against reactor: NRX-A6



- most comprehensively documented of NERVA
- ca. 1150MW continuously for >1h (longest run of NERVA)
- Lowest corrosive mass loss by far (improved fuel coating)
- Highest propulsive efficiency of NRX-A series (ISP=869s)

Chapter 2: Overall Workflow

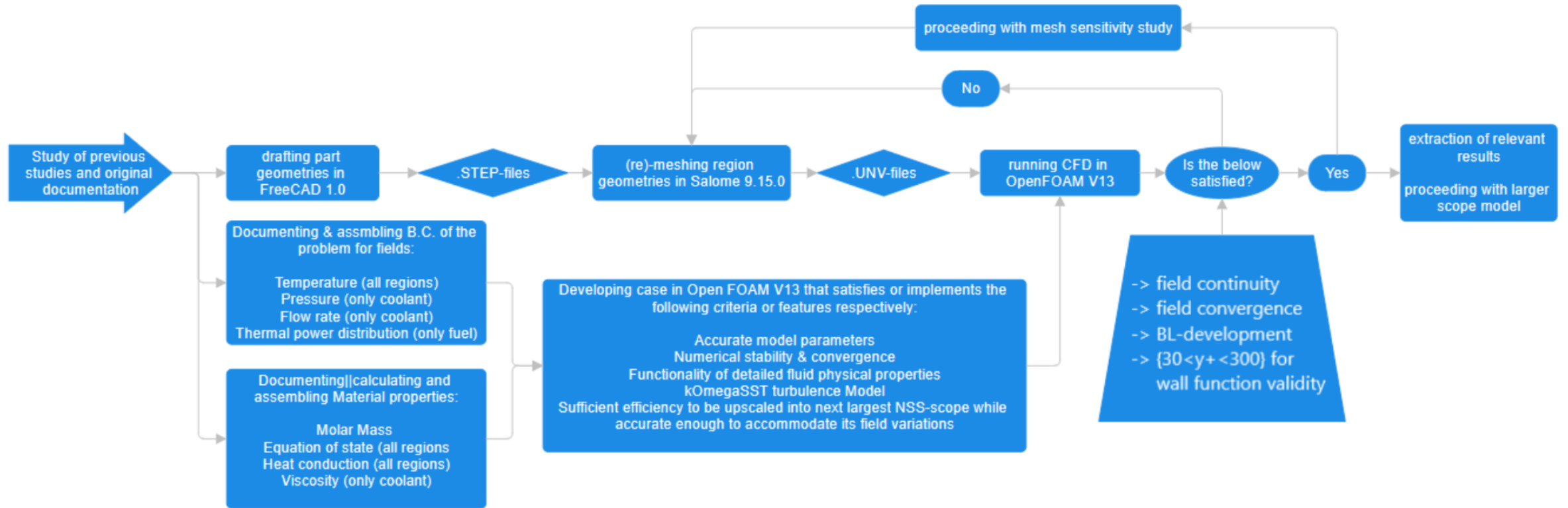


Figure 2.1: Workflow for CFD Model Development

Chapter 3: Theory Review built around PIMPLE algorithm

PIMPLE flowchart in Thesis:

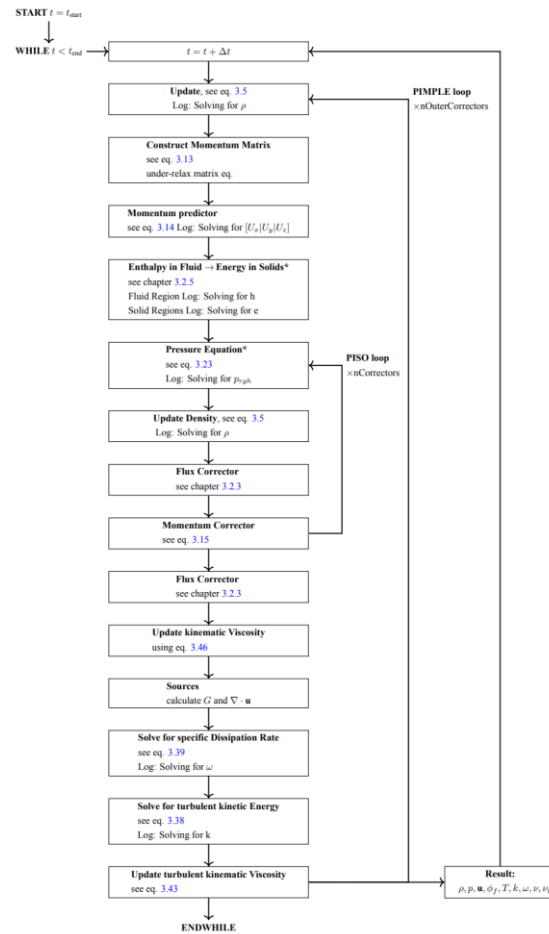


Figure 3.1: Flowchart of the PIMPLE algorithm for this use case.

Typical log.foamMultiRun entry:

```
Coolant_Region Courant Number mean: 0.450157 max: 0.967255
deltaT = 2e-06
Time = 16s

Coolant_Region diagonal: Solving for rho, Initial residual = 0, Final residual = 0, No Iterations 0
Coolant_Region DILUPBiCGStab: Solving for Ux, Initial residual = 8.08331e-11, Final residual = 5.48205e-13, No Iterations 1
Coolant_Region DILUPBiCGStab: Solving for Uy, Initial residual = 7.90428e-11, Final residual = 5.4436e-13, No Iterations 1
Coolant_Region DILUPBiCGStab: Solving for Uz, Initial residual = 2.0115e-13, Final residual = 2.12851e-15, No Iterations 1
Coolant_Region DILUPBiCGStab: Solving for h, Initial residual = 2.57813e-13, Final residual = 2.71544e-15, No Iterations 1
Fuel_Region smoothSolver: Solving for e, Initial residual = 1.06959e-12, Final residual = 1.5478e-16, No Iterations 1
Support_Region smoothSolver: Solving for e, Initial residual = 2.96985e-10, Final residual = 3.57391e-14, No Iterations 1
Coolant_Region DICPBiCGStab: Solving for p_rgh, Initial residual = 1.72733e-12, Final residual = 1.49634e-12, No Iterations 1
Coolant_Region GAMG: Solving for p_rgh, Initial residual = 1.65881e-12, Final residual = 5.80141e-13, No Iterations 1
Coolant_Region diagonal: Solving for rho, Initial residual = 0, Final residual = 0, No Iterations 0
Coolant_Region time step continuity errors : sum local = 4.11952e-14, global = 5.0761e-17, cumulative = 1.01185e-09
Coolant_Region diagonalPBiCG: Solving for omega, Initial residual = 5.94718e-14, Final residual = 9.641e-15, No Iterations 1
Coolant_Region diagonalPBiCG: Solving for k, Initial residual = 2.98206e-13, Final residual = 2.68979e-14, No Iterations 1
ExecutionTime = 111387 s ClockTime = 114999 s
```

Alterations compared to literature examples:

- Regular PIMPLE and RANS flowcharts unified
- Repetition loops matched to corrector settings in fvSolutions
- Order adapted based on log above
- Hyperlinks to theoretical deduction added to enable troubleshooting of physics

Theory review does not include:

- Outlining specific implementation of numerical schemes used
- Discretization of turbulence fields

*Repeated: ×nNonOrthogonalCorrectors

Chapter 5: Model Parameters - Geometry



Figure 5.1: Equivalent flow channel concept - Source: [3, p.119]

Table 5.1: Axial extents of the modeled reactor regions

Region	z-Length	Information
Fuel Region	1320.8 mm (52 in)	spans from cold inlet to hot end —
Support Region	38.1 mm (1.5 in)	spans from fuel hot end to exhaust chamber —
Propellant / Coolant Region	1358.9 mm (53.5 in)	spans coaxially through Fuel & Support

Chapter 5: Model Parameters – Fluid Meshing

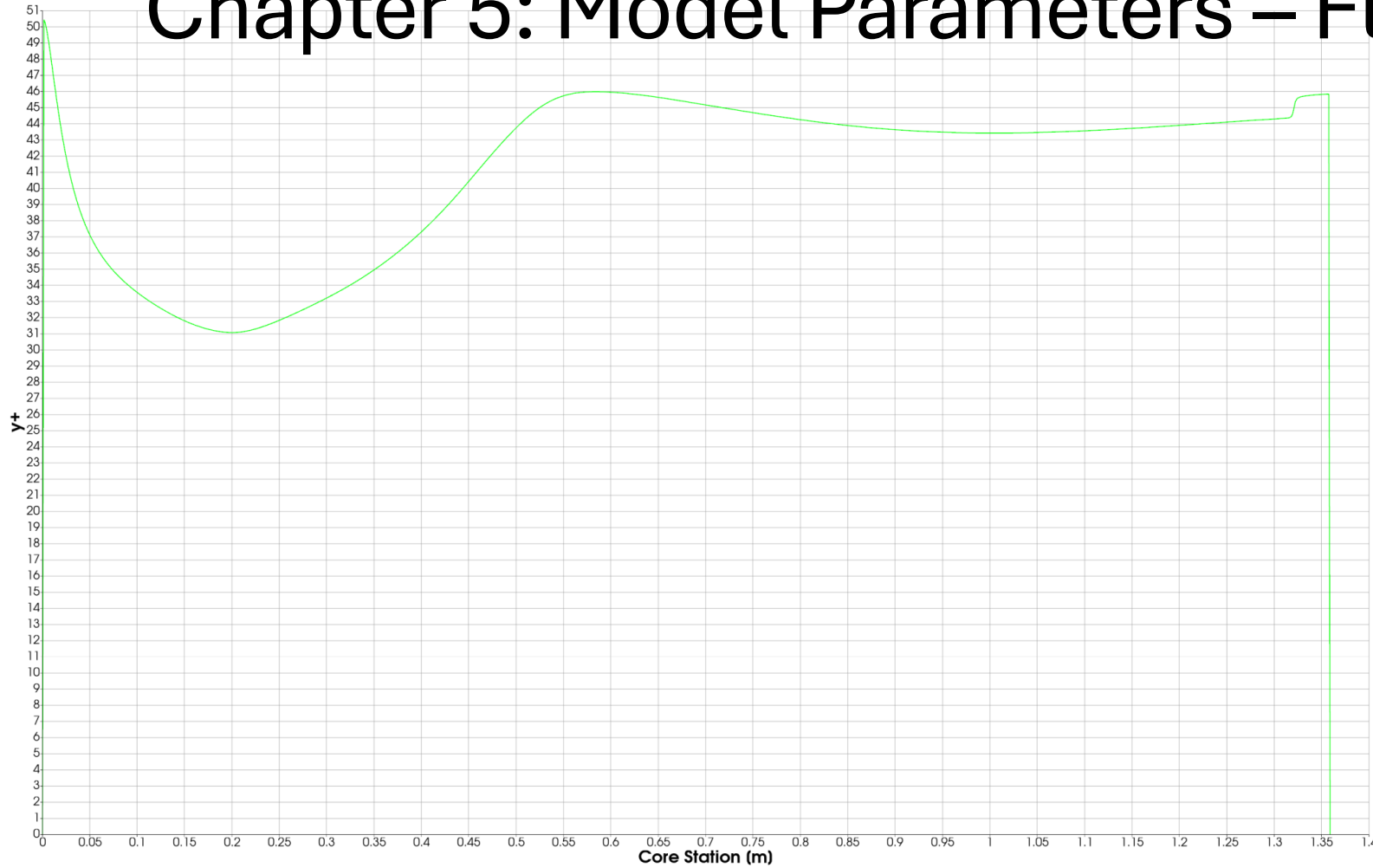


Figure 6.10: Axial distribution of y^+ along the fluid-solid interface

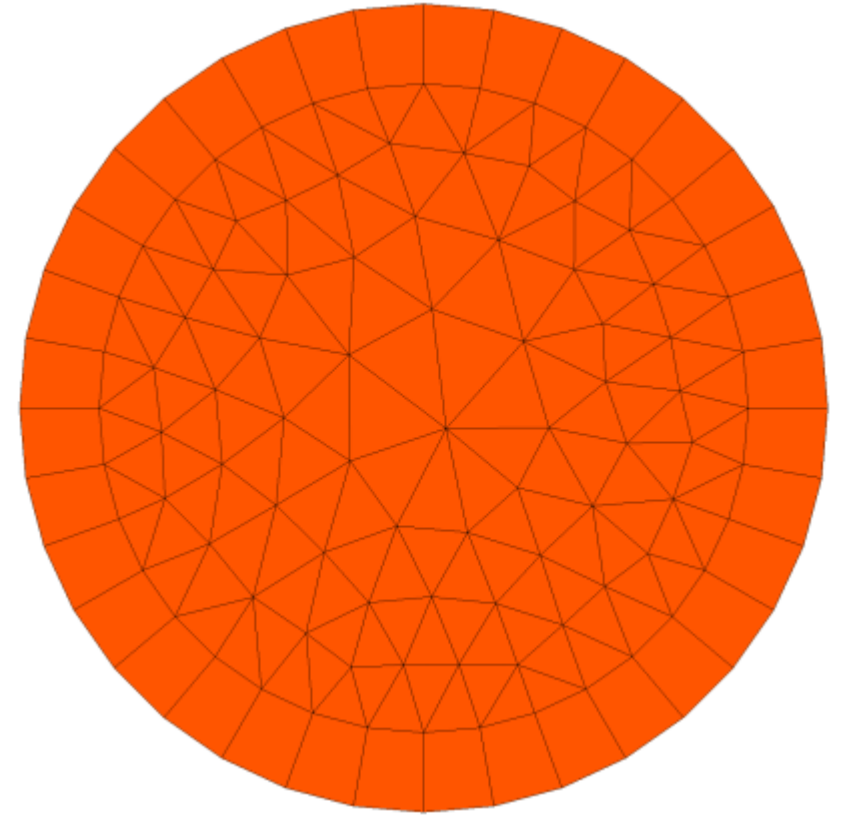
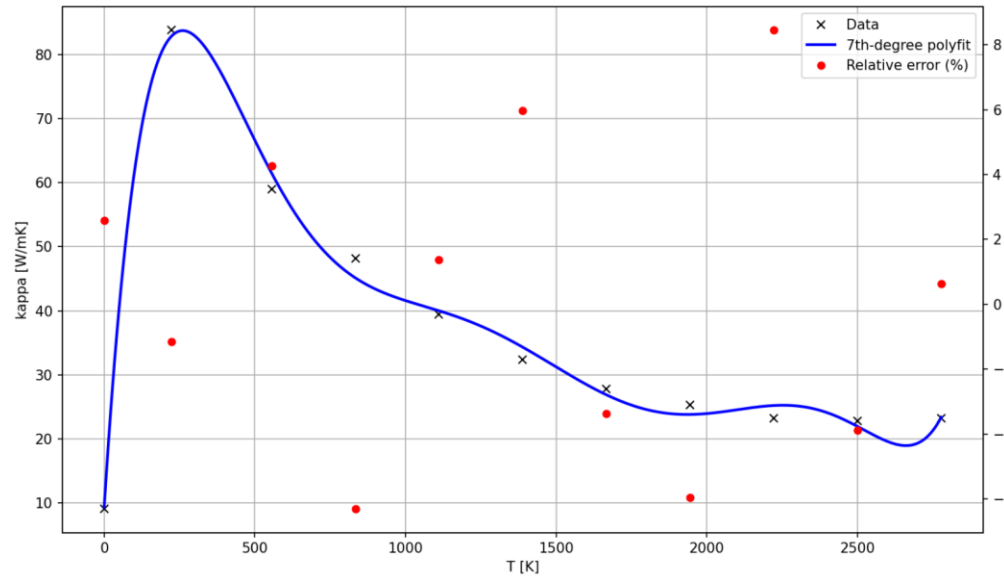


Fig. 5.4 (d) Propellant region mesh used

Why no cross-sectional mesh sensitivity study:

- y^+ margin minimal to begin with
- Enlarged viscous layer would compromise cell size continuity & overall bulk size
- Bulk meshing algorithm difficult to calibrate for such a constrained cross-section

Chapter 5: Model Parameters – Fuel $\kappa(T)$



$$\begin{aligned} \kappa(T) = & 3.19935732 \cdot 10^{-20} T^7 \\ & - 3.42194774 \cdot 10^{-16} T^6 \\ & + 1.47930845 \cdot 10^{-12} T^5 \\ & - 3.30794303 \cdot 10^{-9} T^4 \\ & + 4.05518685 \cdot 10^{-6} T^3 \\ & - 2.63206744 \cdot 10^{-3} T^2 \\ & + 7.47937033 \cdot 10^{-1} T \\ & + 9.37349442 \end{aligned}$$

Figure 5.6: Polynomial fit for fuel element thermal conductivity

- Experimental data only for small high-temperature envelope for one fuel content
- Confidence in data for low-temperature behavior is low (only model estimates)
- Powder quality for graphite extrusion was inconsistent during NRX-A6 fuel production
- Considerable relative error in polynomial fit

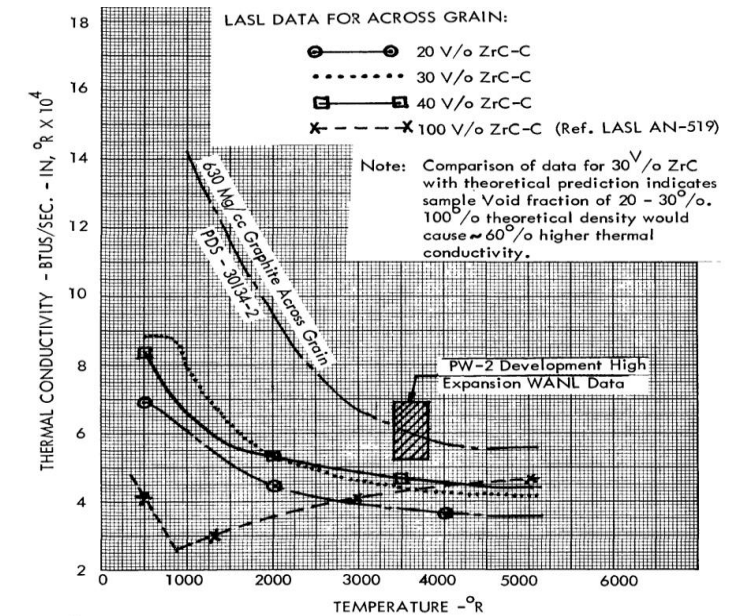


Figure A-1.4.1-10 (CRD) Thermal Conductivity Versus Temperature

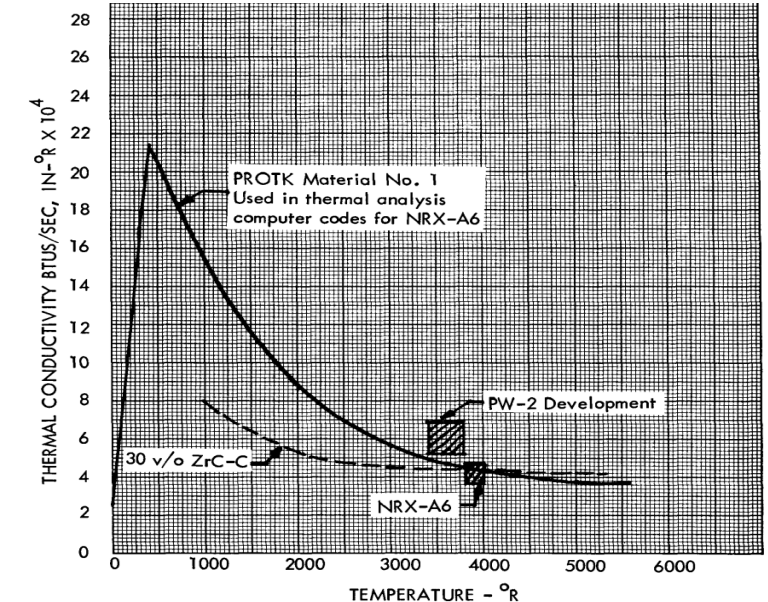
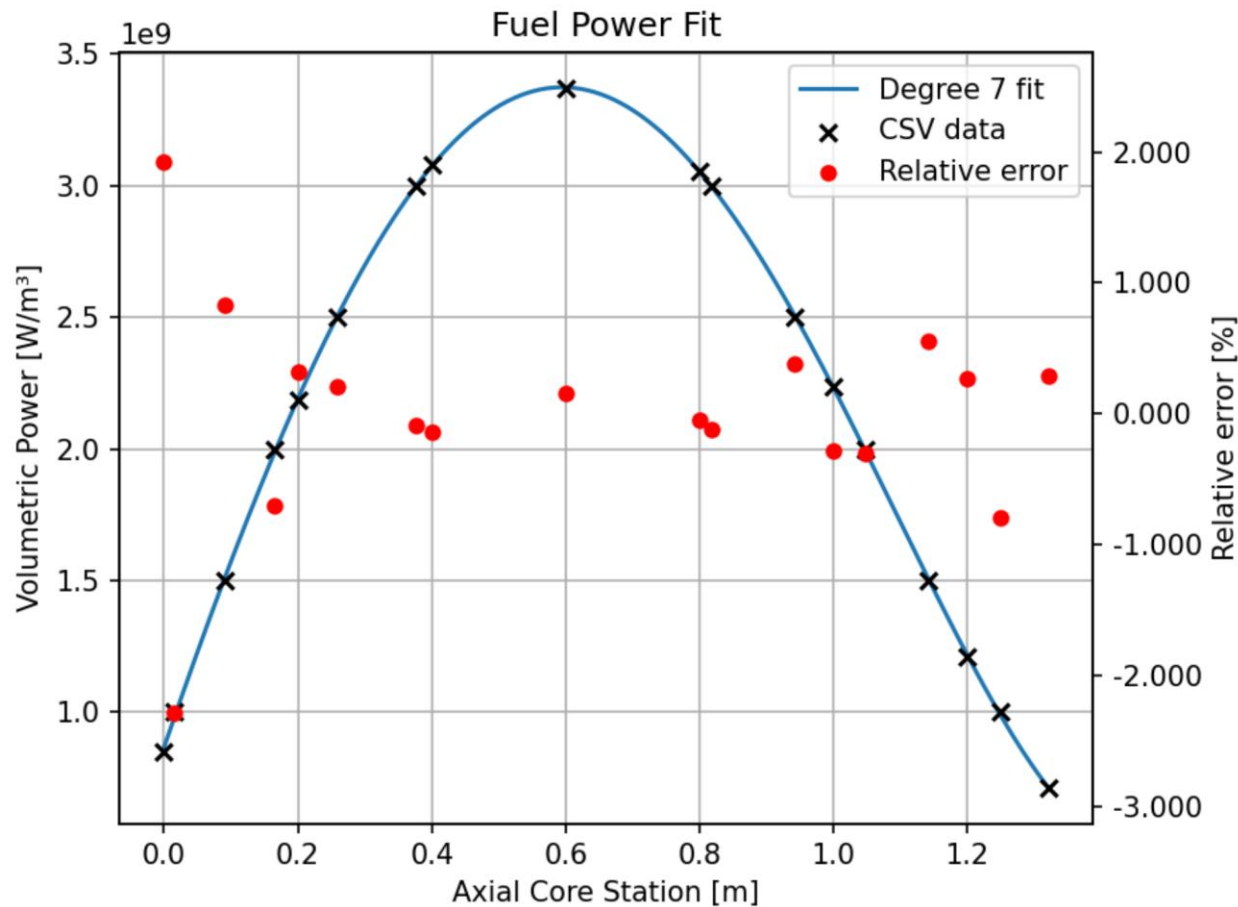


FIGURE A-1.4.1-11 COMPARISON OF 30 V/o COMPOSITE WITH NRX-A6 GRAPHITE THERMAL CONDUCTIVITY. (U)

Chapter 5: Model Parameters – Axial Power Distribution



PROGRAM ELM

```
CC  
CC THIS PROGRAM PERFORMS AEROTHERMODYNAMIC ANALYSIS ON H2 GAS COOLED  
CC NUCLEAR REACTOR CORE FUEL ELEMENTS. THE ANALYSIS IS BASED ON  
CC CONSERVATION OF ENERGY EQUATIONS WITH EXTENSIVE USE OF CORRELATIONS  
CC FOR THE CONVECTIVE HEAT TRANSFER COEFFICIENT, FRICTION FACTOR, AND  
CC ELEMENT AXIAL POWER DISTRIBUTION. THE ANALYSIS IS FOR AN AVERAGE  
CC COOLANT CHANNEL, SUCH AS AT A POINT IN THE CORE WHERE THE RADIAL  
CC POWER OVER AVERAGE POWER RATIO IS UNITY.
```

Figure 5.11: Description of axial power profile noted on the first lines of Walton's code

Why scaling of axial power distribution based on radial position required:

- Average power cluster likely does not contain thermocouples
- Hot channel in center region provides least favorable conditions thermally and numerically
- Cross-sectional gradients in radial direction minimal in center region
- Taking radially innermost reactor temperature measurements allows for most accurate and comprehensive validation

Chapter 5: Model Parameters – Radial Power Scaling

Figure 5.13:

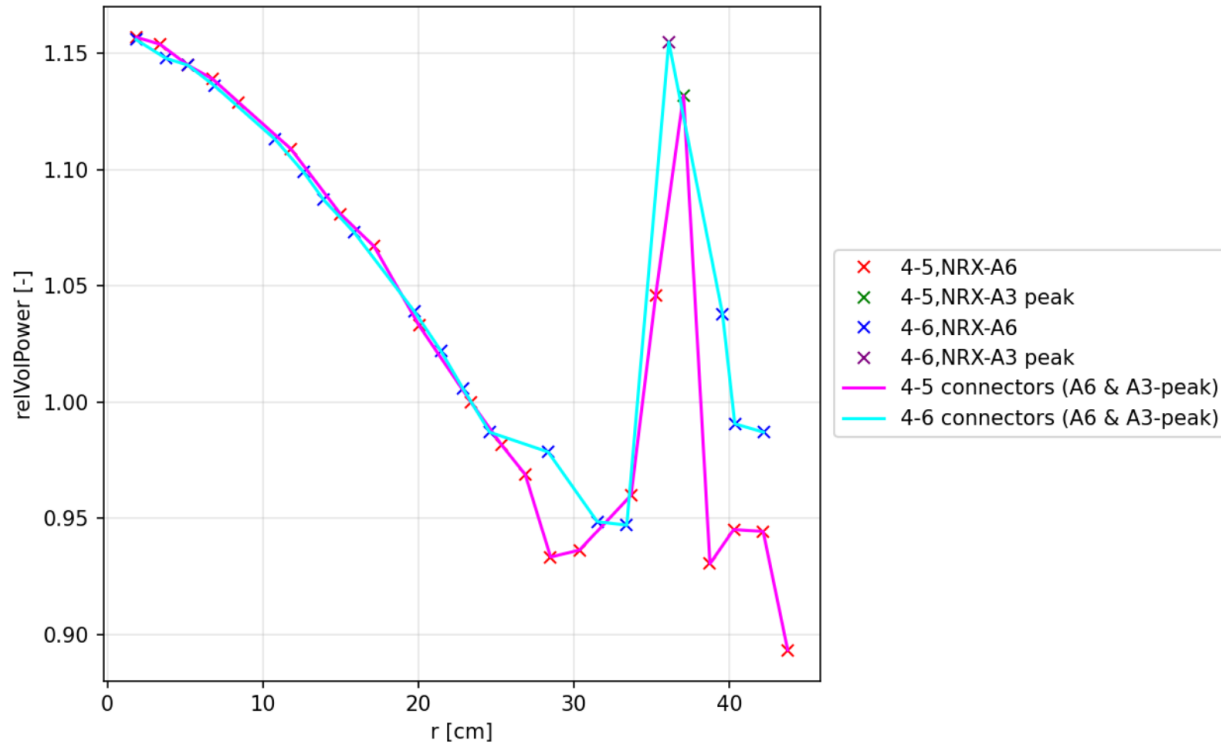
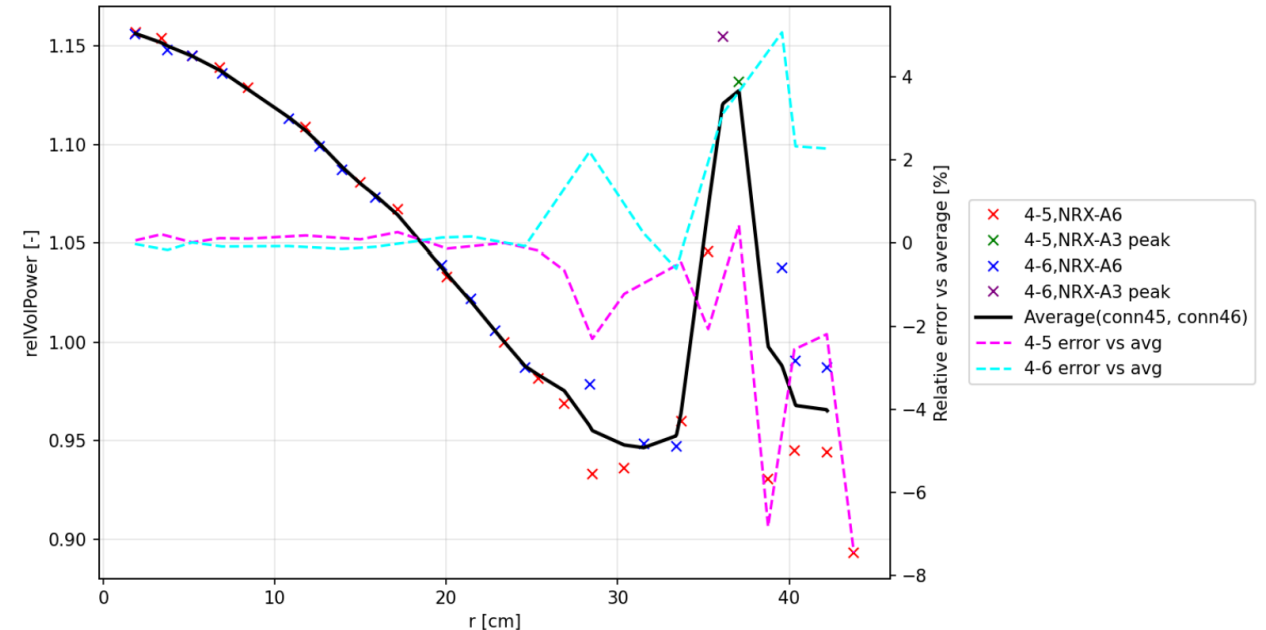


Figure 5.14:



Why this specific method was chosen:

- Considerable angular variability only at core periphery
- Connector averaging instead of points averaging:
Data point radii vary between azimuths

Model Caveats:

- Based on predictive models
- Identical axial power profile assumed across reactor core
- Peripheral peak values historically inferred from other reactors
- Angular power variability at core periphery is smeared
- Small underestimation of peripheral peak

Chapter 6: Validation Results

Temperature:

Model Probe	Axial	OpenFOAMv13	Experiment	Thermocouple	Appendix
Region	Position	[K]	[K]	Location	Source
Fuel_to_Neighbour	2.54cm (1in)	538.7	324 ± 46	Fuel Element	tab. B.1
Fuel_to_Neighbour	50.8cm (20in)	1788	1986.1 ± 5.6	Unfueled El.	tab. B.1
Fuel_to_Neighbour	66.05cm (26in)	2083	2172 ± 11	Unfueled El.	tab. B.2
Outlet_Boundary	135.89cm	2544 ± 2.6	2422 ± 97	Cluster Exit Gas	tab. B.1

Table 6.1: Comparison of regions and positions between OpenFOAM simulation and experimental data.

Source of Value	Region	Axial Position	Temperature [K]
Calculated Curve	Fuel Element	50.8cm (20in)	1720
OpenFoam v13	Fuel (Station 20)	50.8cm (20in)	1788
Calculated Curve	Fuel Element	66.05cm (26in)	2027.8
OpenFoam v13	Fuel (Station 26)	66.05cm (26in)	2083
Calculated Curve	Fuel axial Peak	116.84cm (46in)	2544.4
Calculated Curve	Unfueled El. ax. Peak	111.76cm (44in)	2583.3
OpenFoam v13	Fuel Region ax. Peak	124.14cm	2640.4

Table 6.2: Comparison with Temperature calculated curves in fig. [B.12](#) for measured core stations and peak values

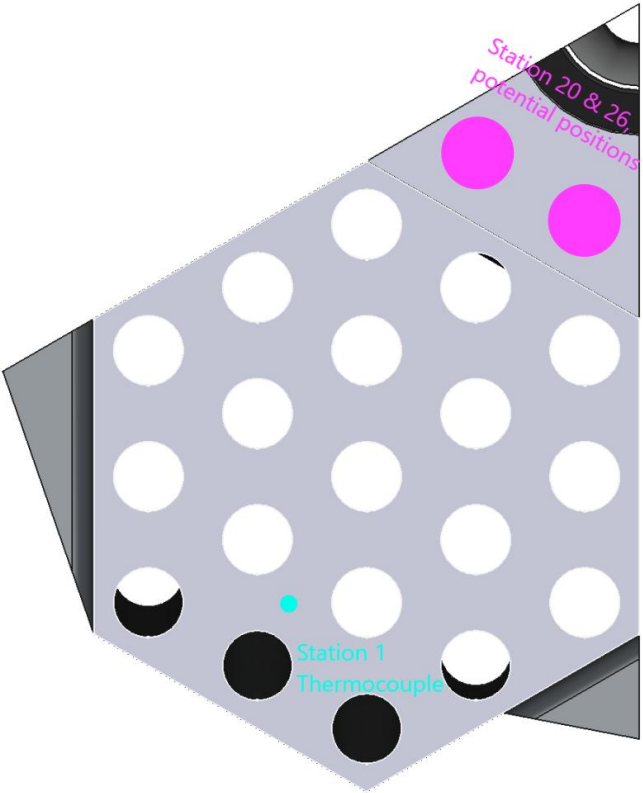
- Fuel temperature within plausible range except for station 1, fully accurate validation requires larger scope (see right)
- Cluster outlet gas temperature shows good agreement (also see appendix table B.1)

Pressure:

Position	OpenFOAM v13	Experiment
Inlet Pressure	4.499MPa	4.889MPa
Outlet Pressure (fixed)	4.033MPa	4.033MPa
Pressure Drop	0.466MPa	0.856MPa

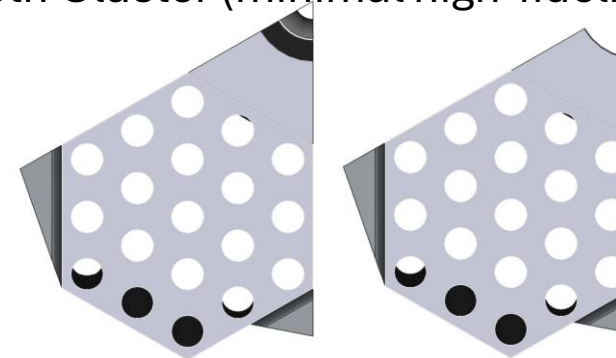
Underestimation

- Likely due to inlet & especially outlet effects
- Additional potential contributors:
- Simplified EoS (In Density graph: $Z=1.044$)
- Potential turbulence model error



Chapter 7 & 8: Application and Conclusion

1/6th Cluster (minimal high-fidelity validation model)



Full Cluster

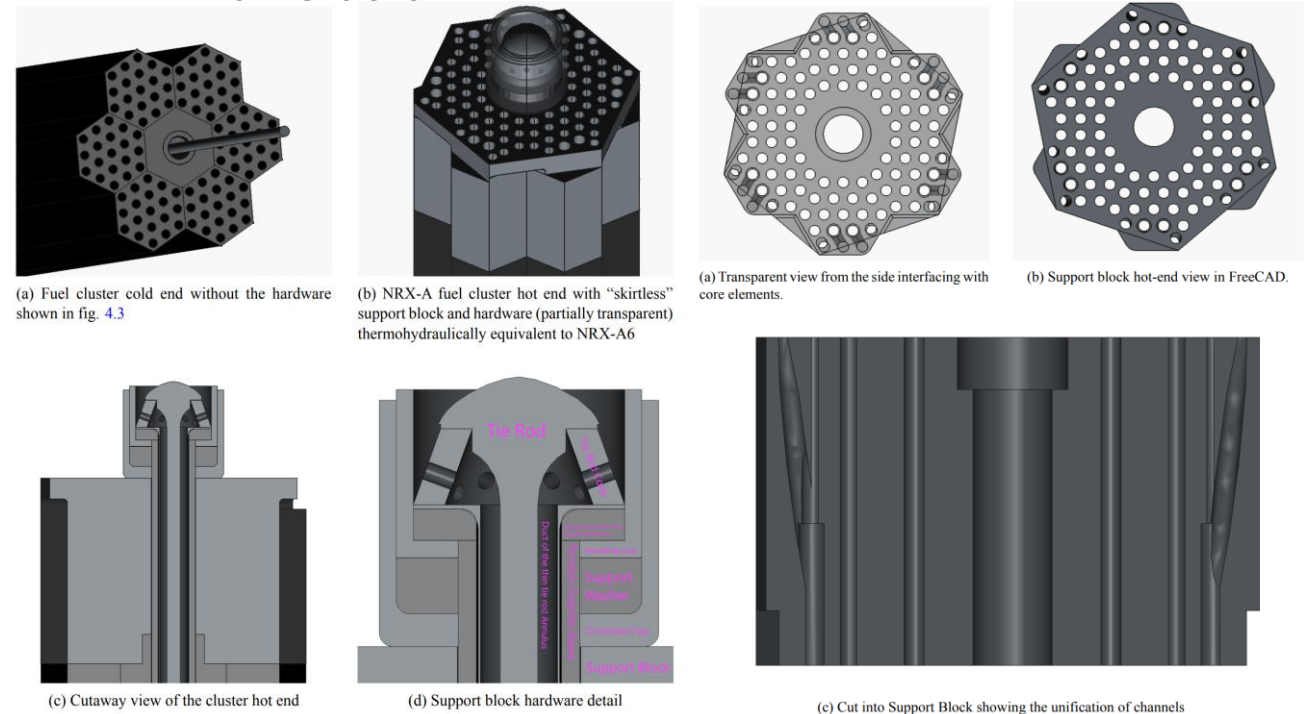


Figure 4.2: Reconstructed NRX-A series fuel cluster geometry rendered in FreeCAD, showing cold-end and hot-end configurations.

Figure C.8: Approximate reconstruction of NRX-A6 support block geometry rendered in FreeCAD. Not all edges could be filleted due to artifacts arising at intersections.

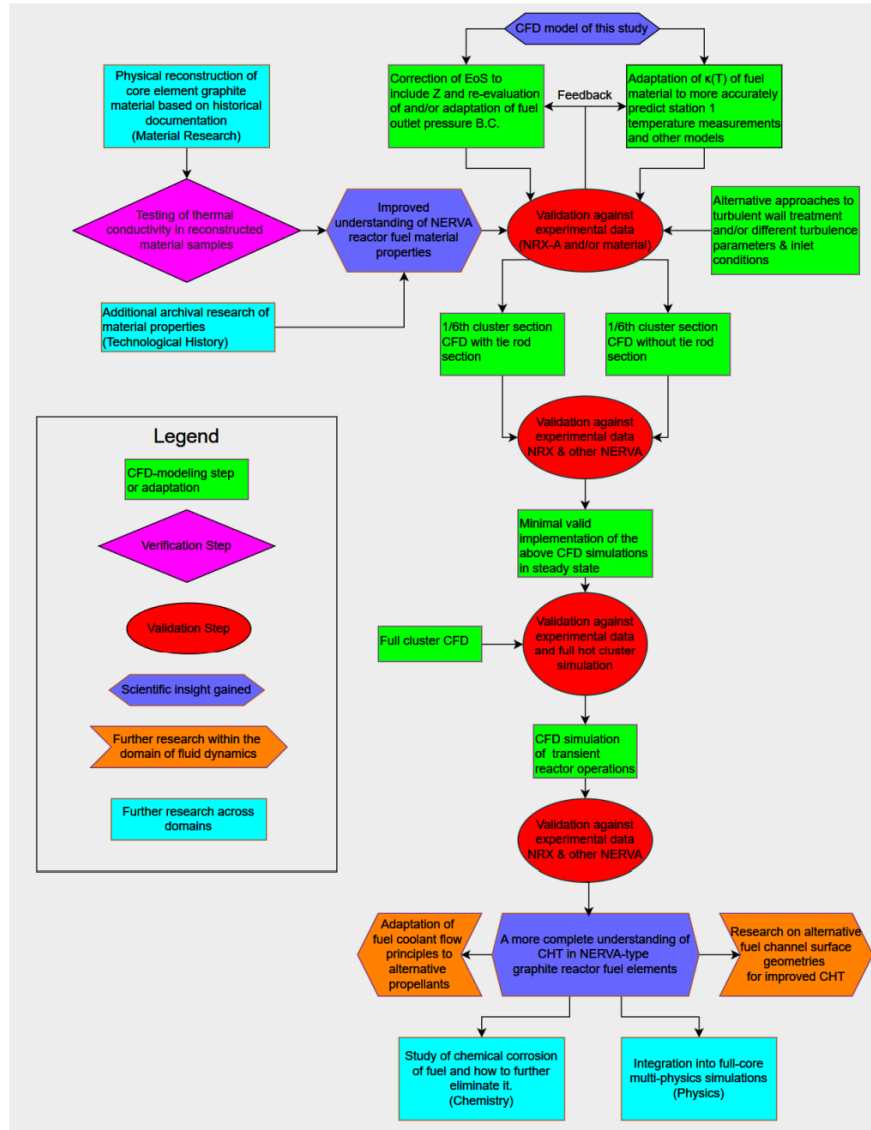


Figure 8.1: Potential research paths based on the findings of this study